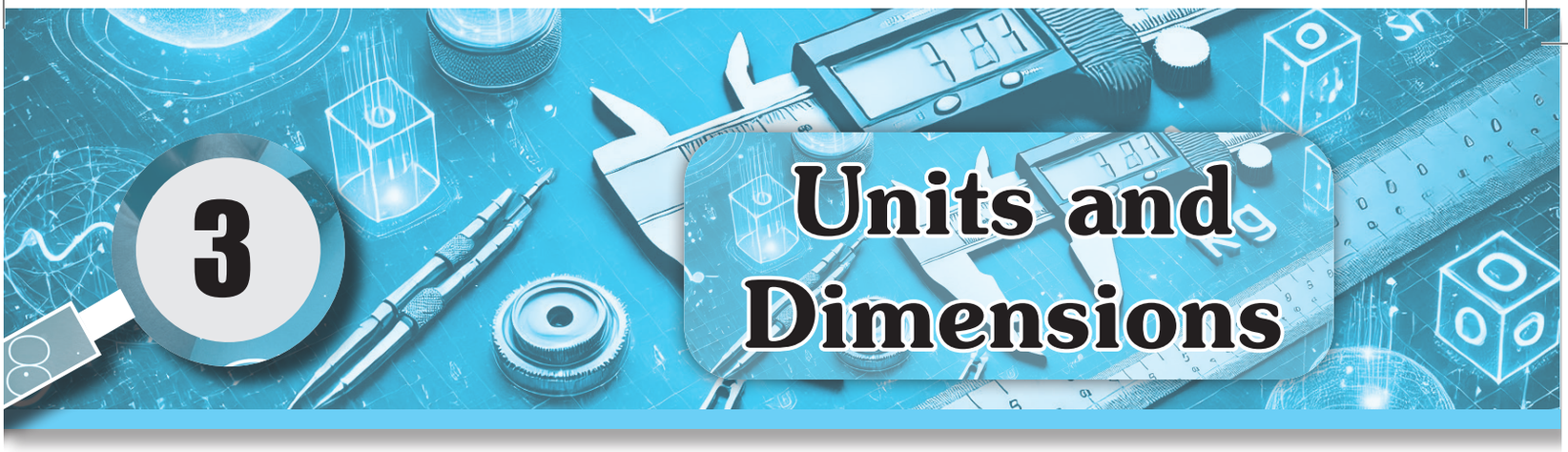


3 Units and Dimensions

Physical Quantity

1. Which of the following group of physical quantity can be considered as a group of fundamental physical quantity.
   (*a*) Force, mass time
   (*b*) Mass, force, acceleration
   (*c*) Velocity, time, displacement
   (*d*) Velocity, momentum, mass
   (*e*) None of the above
2. Which does not have same unit as other:-
   (*a*) watt - sec (*b*) kilowatt - hour
   (*c*) eV (*d*) Joule - sec
3. Unit of distance is :
   (*a*) Femtometer (*b*) Angstrom
   (*c*) Parsec (*d*) Light year
   (*e*) All of these
4. **Assertion (A):** Astronomical unit, light year and parsec measures distance
   Reason (R): Each has dimension of distance.
   (*a*) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).
   (*b*) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).
   (*c*) If Assertion (A) is True but the Reason (R) is False.
   (*d*) If both Assertion (A) & Reason (R) are false.
5. Which of the following is a characteristic of unit?
   (*a*) The unit must be universally accepted
   (*b*) It must be invariable and well defined
   (*c*) It must be of suitable size and easily available
   (*d*) All the above
6. Ratio of two similar physical quantity has units.
   (*a*) True (*b*) False

Unit Conversion

7. The magnitude of physical quantity and units are directly proportional to each other.
   (*a*) True (*b*) False
8. If Area of object is $5m^2$ then find its value in C.G.S unit??
   (*a*) 5×10^4 cm^2 (*b*) 3×10^4 m^2
   (*c*) 2×10^4 cm^2 (*d*) 1×10^4 m^2
9. Convert 25 m/s in C.G.S unit
10. If unit of length becomes double then value of area $5m^2$ in new unit will be :-
    (*a*) $\frac{1}{4}$ (*b*) $\frac{3}{2}$ (*c*) $\frac{5}{4}$ (*d*) $\frac{5}{2}$
11. Convert 1 newton into Dyne.
    (*a*) 10^4 (*b*) 10^5 (*c*) 10^3 (*d*) 10^7
12. Convert values of 10 joule in CGS unit?
    (*a*) 10×10^7 (*b*) 1×10^7 (*c*) 1×10^5 (*d*) 3×10^5
13. In a new system of units, unit of mass is x kg, unit of length is y metre and unit of time is z second. Now if 1 newton = F new units then F =
    (*a*) $\frac{z}{xy}$ (*b*) $\frac{z^2}{xy}$ (*c*) $\frac{z}{xy^2}$ (*d*) $\frac{z}{x^2y}$
14. In new system of unit, unit of length is 10 m, unit of time is 2s, unit of mass is 5 kg, then find unit of torque in new system of unit.
    (*a*) 125 Nm (*b*) 0.125 Nm
    (*c*) 8 Nm (*d*) 8×10^{-3} Nm
15. If unit of length 10 m and unit of mass is 5 kg and unit of time is 2 sec then, find value of 10 Joule energy in new system of unit.
    (*a*) $\frac{2}{25}$ (*b*) $\frac{3}{8}$ (*c*) $\frac{1}{16}$ (*d*) $\frac{2}{3}$
16. If unit of mass becomes double, unit of length becomes half and unit of time becomes $\frac{1}{4}$ th then find value of 10 N force in new system of unit.
17. The density of a material in CGS system of units is 4 g cm^{-3} In a system of units in which unit of length is 10 cm and unit of mass is 100 g, the value of density of material will be
    (*a*) 0.04 (*b*) 0.4 (*c*) 40 (*d*) 400
18. If unit of length 10 m and unit of mass is 5 kg and unit of time is 2 sec then, find unit of energy in new system.
    (*a*) 25 kg.m^2/s^2 (*b*) 125 kg.m^2/s^2
    (*c*) 1 kg.m^2/s^2 (*d*) 108 kg.m^2/s^2
19. If unit of mass becomes double, unit of length becomes half and unit of time becomes $\frac{1}{4}$ th then find unit of force.
    (*a*) 10 kg.m^2/s^2 (*b*) 25 kg.m^2/s^2
    (*c*) 1 kg.m^2/s^2 (*d*) 16 kg.m^2/s^2
20. In a new system of units, unit of mass is α kg unit of length is β m and unit of time is γ s. In this system, 10J will be represented as.
    (*a*) $10\alpha^{-1}\beta^2\gamma^2$ (*b*) $10\alpha^{-2}\beta^{-1}\gamma^{-2}$
    (*c*) $10\alpha^{-1}\beta^{-2}\gamma^2$ (*d*) $10\alpha\beta^2\gamma^{-2}$

21. Given below are two statements:

Statement I: Two physical quantities having same dimensions, may have different units.

Statement II: Shake and light year, both measure time.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both statement I and statement II are correct.

(*b*) Statement I is correct and statement II is incorrect

(*c*) Statement I is incorrect and statement II is correct

(*d*) Both statements I and statements II are incorrect

22. Light year is used to measure:

(*a*) distance between stars

(*b*) distance between atoms

(*c*) revolution time of earth around sun

(*d*) none of these

23. The wrong unit conversion among the following is :

(*a*) 1 angstrom = 10^{-10} m

(*b*) 1 fermi = 10^{-15} m

(*c*) 1 light year = 9.46×10^{15} m

(*d*) 1 astronomical unit = 1.496×10^{-11} m

24. The unit for nuclear dose given to a patient is :

(*a*) fermi (*b*) rutherford

(*c*) curie (*d*) roentgen

25. Which of the following is not the unit of time ?

(*a*) microsecond (*b*) leap year

(*c*) lunar months (*d*) parallactic second

(*e*) Solar day

26. The unit of Stefan's constant σ is :

(If rate of heat radiation is given by σAT^4 where *A* is Area and T is temperature)

(*a*) $Wm^{-2}K^{-1}$ (*b*) $Wm^2 K^{-4}$

(*c*) $Wm^{-2}K^{-4}$ (*d*) $Wm^{-2} K^4$

27. Match Column-I with Column-II and select correct option.

Column-I		Column-II	
(A)	Mega	(P)	10^{-9}
(B)	Nano	(Q)	10^{-15}
(C)	Micro	(R)	10^{6}
(D)	Femto	(S)	10^{-6}
(E)	Pico	(T)	10^{-12}

	A	B	C	D	E
(*a*)	R	P	S	T	Q
(*b*)	S	P	R	Q	T
(*c*)	R	P	S	Q	T
(*d*)	S	P	R	T	Q

28. Two clocks are being tested against a standard clock located in a national laboratory. At 12 : 00 : 00 noon by the standard clock, the readings of the two clocks are :

Days	Clock 1	Clock 2
Monday	12:00:05	10:15:06
Tuesday	12:01:15	10:14:59
Wednesday	11:59:08	10:15:18
Thursday	12:01:50	10:15:07
Friday	11:59:15	10:14:53
Saturday	12:01:30	10:15:24
Sunday	12:01:19	10:15:11

If you are doing an experiment that requires precision time interval measurements, which of the two clocks will you prefer?

29. Which of the following ratios express pressure ?

(i) Force/Area

(ii) Energy/Volume

(iii) Energy/Area

(iv) Force/Volume

(*a*) Only (i) is correct. (*b*) Only (ii) is correct.

(*c*) Only (iii) is correct. (*d*) Both (i) and (ii) are correct.

30. The unit of thermal conductivity is :

(*a*) $Wm^{-1}K^{-1}$ (*b*) $Hm\ K^{-1}$

(*c*) $Jm^{-1}\ K^{-1}$ (*d*) $Wm\ K^{-1}$

Dimension

31. The dimension of mutual inductance is:

(*a*) $[ML^2T^{-2}\ A^{-1}]$ (*b*) $[ML^2T^{-3}\ A^{-1}]$

(*c*) $[ML^2T^{-2}\ A^{-2}]$ (*d*) $[ML^2T^{-3}\ A^{-2}]$

32. The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be

(*a*) $[ML^{-1}T^{-1}]$ (*b*) $[ML^{-1}T^{-2}]$

(*c*) $[ML^2T^{-1}]$ (*d*) $[ML^{-1}L^3\ T^0]$

33. Dimension of Stress

34. Dimension of electric resistance

35. Dimension of Self Induction

36. Dimension of Permeability

37. Dimension of Magnetic field and magnetic flux

38. Dimension of Coefficient of viscosity

39. Dimension of $\frac{1}{\mu_0 \in_0}$ should be equal to

(*a*) T^2/L^2 (*b*) T/L

(*c*) L^2/T^2 (*d*) L/T

40. If the dimensions of a physical quantity are given by $M^aL^bT^c$, then the physical quantity will be

(*a*) Velocity if $a = 1, b = 0, c = -1$

(*b*) Acceleration if $a = 1, b = 0, c = -2$

(*c*) Force if $a = 0, b = -1, c = -2$

(*d*) Pressure if $a = 1, b = -1, c = -2$

41. If E, L, M and G denote the quantities as energy, angular momentum, mass and constant of gravitation respectively, then the dimensions of *P* in the formula $P = EL^2M^{-5}G^{-2}$ are:

(*a*) $[M^0L^0T^0]$ (*b*) $[M^1L^1T^{-2}]$

(*c*) $[M^0L^1T^0]$ (*d*) $[M^{-1}L^{-1}T^2]$

42. A quantity f is given by $f = \sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is that of

(*a*) Energy (*b*) Momentum

(*c*) Area (*d*) Volume

43. Which two of the following five physical parameters have the same dimensions?

44. Energy density 2. Refractive index

45. Dielectric constant 4. Young's modulus

46. Magnetic field

(*a*) 1 and 4 (*b*) 1 and 5

(*c*) 2 and 4 (*d*) 3 and 5

44. Which of the following pairs have same dimensional formula?

(*a*) Bulk modulus and energy density

(*b*) Latent heat and velocity

(*c*) Specific heat and latent heat

(*d*) Both (*a*) and (*b*)

45. Which pair do not have equal dimensions?

(*a*) Energy and torque

(*b*) force and impulse

(*c*) Angular momentum and Planck's constant

(*d*) Elastic modulus and pressure

46. The unit of electric flux is

(*a*) $\frac{Vm^2}{C}$ (*b*) volt-second

(*c*) $\frac{Nm^2}{C}$ (*d*) $\frac{N}{C^2\ m}$

47. Match List-I with List-II:

List-I		List-II	
A.	Surface tension	I.	$kg\ m^{+1}s^{-1}$
B.	Pressure	II.	$kg\ m^{-1}s^{-2}$
C.	Viscosity	III.	$kg\ m^{-1}s^{-2}$
D.	Impulse	IV.	$kg\ s^{-2}$

Choose the correct answer from the options given below:

(*a*) A-IV, B-III, C-II, D-I

(*b*) A-IV, B-III, C-I, D-II

(*c*) A-III, B-IV, C-I, D-II

(*d*) A-II, B-I, C-III, D-IV

48. Match List-I with List-II:

List-I		List-II	
A.	Torque	I.	$kg\ m^{-1}s^{-2}$
B.	Energy density	II.	$kg\ ms^{-1}$
C.	Pressure gradient	III.	$kg\ m^{-2}s^{-2}$
D.	Impulse	IV.	$kg\ m^2\ s^{-2}$

Choose the correct answer from the options given below:

(*a*) A-IV, B-III, C-I, D-II (*b*) A-I, B-IV, C-III, D-II

(*c*) A-IV, B-I, C-II, D-III (*d*) A-IV, B-I, C-III, D-II

49. Match List-I with List-II:

List-I		List-II	
A.	Planck's constant (h)	I.	$[M^1L^2T^{-2}]$
B.	Stopping potential (Vs)	II.	$[M^1L^1T^{-1}]$
C.	Work function (ϕ)	III.	$[M^1L^2T^{-1}]$
D.	Momentum (p)	IV.	$[M^1L^2T^{-3}A^{-1}]$

(*a*) A-III, B-I, C-II, D-IV (*b*) A-III, B-IV, C-I, D-II

(*c*) A-II, B-IV, C-III, D-I (*d*) A-I, B-III, C-IV, D-II

50. Match List-I with List-II:

List-I		List-II	
A.	Young's Modulus (Y)	I.	$[M\,L^{-1}T^{-1}]$
B.	Co-efficient of Viscosity (η)	II.	$[M\,L^2T^{-1}]$
C.	Planck's Constant (h)	III.	$[M\,L^{-1}T^{-2}]$
D.	Work Function (ϕ)	IV.	$[M\,L^2T^{-2}]$

Choose the correct answer from the options given below:

(*a*) A-II, B-III, C-IV, D-I (*b*) A-III, B-I, C-II, D-IV

(*c*) A-I, B-III, C-IV, D-II (*d*) A-I, B-II, C-III, D-IV

51. Match List-I with List-II:

List-I (Quantity)		List-II (Dimensional Formula)	
A.	Pressure gradient	I.	$[M^0L^2\,T^{-2}]$
B.	Energy density	II.	$[ML^{-1}T^{-2}]$
C.	Electric field	III.	$[M^1\,L^{-2}T^{-2}]$
D.	Latent heat	IV.	$[M^1\,L^1\,T^{-3}\,A^{-1}]$

Choose the correct answer from the options given below:

(*a*) A-III, B-II, C-I, D-IV (*b*) A.-II, B-III, C-IV, D-I

(*c*) A-III, B-II, C-IV, D-I (*d*) A-II, B-III, C-I, D-IV

52. Match List-I with List-II:

List-I		List-II	
A.	Angular momentum	I.	$[M\,L^2\,T^{-2}]$
B.	Torque	II.	$[ML^{-2}T^{-2}]$
C.	Stress	III.	$[ML^2T^{-1}]$
D.	Pressure gradient	IV.	$[ML^{-1}\,T^{-2}]$

Choose the correct answer from the options given below:

(*a*) A-I, B-IV, C-III, D-II (*b*) A-III, B-I, C-IV, D-II

(*c*) A-II, B-III, C-IV, D-I (*d*) A-IV, B-II, C-I, D-III

53. Match the List-I with List-II.

List-I		List-II	
A.	Coefficient of viscosity	I.	$M\,L^2\,T^{-2}$
B.	Surface tension	II.	$ML^2\,T^{-1}$
C.	Angular momentum	III.	$ML^{-1}T^{-1}$
D.	Rotational kinetic energy	IV.	$ML^0\,T^{-2}$

Choose the correct answer from the options given below :

(*a*) (A)-(II), (B)-(I) , (C)-(IV), (D)-(III)

(*b*) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(*c*) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(*d*) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

54. If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time?

(*a*) RC (*b*) L/R (*c*) $\sqrt{LC}$ (*d*) L/C

55. **Assertion (A):** The dimensional formula for product of resistance and conductance is same as for dielectric constant.

Reason (R): Both have dimensions of time constant

(*a*) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).

(*b*) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).

(*c*) If Assertion (A) is True but the Reason (R) is False.

(*d*) If both Assertion (A) & Reason (R) are false.

56. A quantity x is given by (IF $v^{2/}$ WL^4) in terms of moment of inertia I, force F, velocity v , work W and length L . The dimensional formula for x is same as that of

(*a*) Coefficient of viscosity (*b*) Force constant

(*c*) Energy density (*d*) Planck's constant

57. The pair of quantities having same dimensions is

(*a*) Impulse and Surface Tension

(*b*) Angular momentum and Work

(*c*) Work and Torque

(*d*) Young's modulus and Energy

58. Which two of the following five physical parameters have the same dimensions ?

(A) energy density (B) refractive index

(C) dielectric constant (D) Young's modulus

(E) magnetic field

(*a*) A and D (*b*) A and E

(*c*) B and D (*d*) C and E

59. Choose the incorrect statement

(*a*) Solid angle has a unit but no dimensions

(*b*) Relative density has neither unit nor dimensions

(*c*) Universal gravitational constant (G) has a SI unit $Nm^2 kg^{-2}$

(*d*) Reynolds number is having a unit

60. If y = force and x = velocity then dimension of $\frac{dy}{dx}$

(*a*) $[MT^{-1}]$ (*b*) $[LT^{-2}]$

(*c*) $[ML^2T^2]$ (*d*) $[MLT^{-1}]$

61. **Assertion (A):** If x and y are the distances along x and y axes respectively then the dimensions of $\frac{d^3y}{dx^3}$ is $M^0 L^{-2} T^0$.

Reason (R): Dimensions of $\int_a^b y dx$ is $M^0 L^2 T^0$

(*a*) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).

(*b*) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).

(*c*) If Assertion (A) is True but the Reason (R) is False.

(*d*) If both Assertion (A) & Reason (R) are false.

62. Fill in the blanks with correct statement, according to given statement

Dimension	(1) _____	(2) _____	(*c*) A physical quantity have dimension	(*d*) A physical quantity does not have dimension
Unit	(*a*) A physical quantity have unit	(*b*) A physical quantity does not have unit	(3) _____	(4) _____

63. A unitless physical quantity may have dimension

(*a*) True (*b*) False

64. A dimensionless physical quantity may be unitless

(*a*) True (*b*) False

65. A physical quantity have unit must have dimension

(*a*) True (*b*) False

66. A physical quantity have dimension may have unit

(*a*) True (*b*) False

67. Select correct options

(*a*) Two physical quantities of different dimensions may have same unit

(*b*) Two physical quantities of different units may have same dimensions

(*c*) Unit less quantities must be dimensionless /

(*d*) Both (*b*) & (*c*)

68. **Assertion (A):** The unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$ have units of distance and dimensions $[M^0L^1T^0]$

Reason (R): The product of a scalar and a vector is a new scalar.

(*a*) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).

(*b*) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).

(*c*) If Assertion (A) is True but the Reason (R) is False.

(*d*) If both Assertion (A) & Reason (R) are false.

69. Different physical quantity can have same dimension?

70. Two quantity have different dimension, can be same physical quantity.

71. Which of the following is a dimensional constant?

(*a*) Relative density (*b*) Gravitational constant

(*c*) Refractive index (*d*) Poisson's ratio

72. The relation between [E] and [B] is

(*a*) [E] = [B][L][T] (*b*) $[E] = [B][L]^{-1}[T]$

(*c*) $[E] = [B][L][T]^{-1}$ (*d*) $[E] = [B][L]^{-1}[T]^{-1}$

73. In the expression $P = El^2m^{-5}G^{-2}$, where E, m, l and G denote energy, mass, angular momentum and gravitational constant, respectively. The dimensions of P are

(*a*) $[MLT^0]$ (*b*) $[M^2LT^{-1}]$

(*c*) $[M^0L^0T^0]$ (*d*) $[M^0 LT^{-2}]$

74. The pairs of physical quantities that have the same dimensions is/are

I. Volumetric strain and coefficient of friction.

II. Disintegration constant of a radioactive substance and frequency of light wave.

III. Heat capacity and gravitational potential.

IV. Stefan's constant and Wien's constant.

(*a*) I, II and IV (*b*) I, III and IV

(*c*) I, II and III (*d*) I and II

75. Match the following columns.

Column-I		Column-II	
(A)	A physical quantity which has a unit but no dimensions	1.	Gravitational constant
(B)	A physical quantity which has neither unit nor dimensions	2.	Reynold number
(C)	A constant which has a unit	3.	Strain
(D)	A constant which has no unit	4.	Plane angle

(*a*) 4 3 1 2 (*b*) 3 4 2 1 (*c*) 1 2 3 4 (*d*) 1 4 2 3

Principle of Homogeneity

76. If P, Q, R are physical quantities having different dimensions. Which of following combinations can never be a meaningful quantity?

i. PQ – R ii. PR–Q^2/R

iii. (P–Q)/R iv. (R+Q)/P

(*a*) i, ii, iii, iv (*b*) both iii and iv

(*c*) both i and ii (*d*) i, iii, iv

77. $F = at + bt^2$, Find dimension of a and b

78. V = A sin(Bt) find dimension of A and B

79. $E = Ax + \frac{Bt}{(C - t^2)}$, Find dimension of A, B and C

80. $\left(P + \frac{q}{V^2}\right) \times (b - v^2) = \frac{nRT}{3}$ Find dimension of b?

81. The velocity v of a particle at time t is given by $v = at + \frac{b}{t + c}$, where *a, b* are constants. The dimensions of *a,b* and *c* are

(*a*) [L],[LT] and [LT^{-2}] (*b*) [LT^{-2}],[L] and [T]

(*c*) [L^2],[T] and [LT^{-2}] (*d*) [LT^{-2}],[LT] and [L]

82. An equation is given here $\left(P + \frac{a}{v^2}\right) = b\frac{\theta}{V}$ where *P*= Pressure, *V* = Volume and θ = Absolute temperature. If *a* and *b* are constants, the dimensions of *a* will be

(*a*) [$MT^{-5}T^{-1}$] (*b*) [ML^5T^{-2}]

(*c*) [ML^5T^{-2}] (*d*) [$M^{-1}L^5T^2$]

83. Electric field in a certain region is given by $\vec{E} = \left(\frac{A}{x^2}\hat{i} + \frac{B}{y^3}\hat{j}\right)$. The SI unit of *A* and *B* are:

(*a*) $Nm^3\ C^{-1}$; $N\ m^2C^{-1}$ (*b*) $Nm^2\ C^{-1}$; $Nm^3\ C^{-1}$

(*c*) Nm^3 C; Nm^2 C (*d*) $Nm^3\ C^{-1}$; Nm^3 C

84. In the equation $\left[x + \frac{a}{y^2}\right][y - b] = RT, x$ is pressure, *y* is volume, *R* is universal gas constant & T is temperature. The physical quantity equivalent to ratio $\frac{a}{b}$ is

(*a*) Impulse (*b*) Energy

(*c*) Pressure gradient (*d*) Coefficient of viscosity

85. The time dependence of a physical quantity *p* is given by $p = p_0 \exp(-at^2)$, where *a* is a constant and *t* is the time. The constant *a*

(*a*) is dimensionless (*b*) has dimensions [T^{-2}]

(*c*) has dimensions [T^2] (*d*) has dimensions of *p*

86. **Assertion (A):** Force is not added with pressure
Reason (R): Quantities having different dimension are not added or subtracted.

(*a*) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).

(*b*) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).

(*c*) If Assertion (A) is True but the Reason (R) is False.

(*d*) If both Assertion (A) & Reason (R) are false.

87. If A and B are physical quantity having different dimension then which of the following expression is correct?

(*a*) A + B (*b*) A – B

(*c*) $A - B^2$ (*d*) AB + B

88. Y = A sin (ωt – *kx* + ϕ) find $\dim^n$ of A, ω, k ϕ and where y and x is position.

89. Velocity $V = \frac{\alpha}{\beta}e^{-\alpha t}$ then find dimension of α and β.

90. If velocity $v = \alpha t + \beta x - \gamma xt$, then find dimension of α, β and γ

91. A force is represented by $F = ax^2 + bt^{1/2}$ where *x* = distance and *t* = time. Then The dimensions of b^2/a are

(*a*) $M\ L^3\ T^{-3}$ (*b*) $M\ L\ T^{-2}$

(*c*) $M\ L^{-1}\ T^{-1}$ (*d*) $M\ L^2\ T^{-3}$

92. On the basis of dimension, decide which of the following relation for displacement of a particle is not correct

(*a*) $y = a\sin\left(\frac{2\pi t}{T}\right)$

(*b*) $y = a\sin\left(\frac{vt}{\lambda}\right)$

(*c*) $y = \sqrt{2}a\sin\left(\frac{2\pi t}{T}\right)\cos\left(\frac{2\pi x}{\lambda}\right)$

(*d*) $y = \frac{x}{t}\sin(\lambda t)$

93. Velocity $V = A\sin\left(x\sqrt{\alpha t}\right)$, find dimension of α

94. Velocity $V = \frac{\alpha}{\beta + \sqrt{\text{density}}}$, find dimension of α and β.

95. $\int \frac{dx}{\sqrt{2ax - x^2}} = b$; find dimension of *a* and *b* of *x* is distance

96. Momentum of object is given as $P = \alpha t\,[1 - \beta^{\gamma x}]$ then, find dimension of α, β and γ. Where t is time *x* is position.

97. The equation of a circle is given by $x^2 + y^2 = a^2$, where *a* is the radius. If the equation is modified to change the origin other than (0, 0), then find out the correct dimensions of A and B in a new equation: $(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$. The dimensions of ties given as [T^{-1}]

(*a*) A = [$L^{-1}T$], B = [LT^{-1}] (*b*) A = [LT], B = [$L^{-1}T^{-1}$]

(*c*) A = [$L^{-1}T^{-1}$], B = [LT^{-1}] (*d*) A = [$L^{-1}\ L^{-1}$], B = [LT]

98. Position of object Y is given $Y = A\sin(kx - \omega t)$ then, find dimension of $k; \omega$ and A .

99. If $Y = A\sin\left[\frac{2\pi}{\lambda}(ct - x)\right]$. Find dimension of c.

100. Equation of force $F = \frac{a - x}{bt^2}$ then find dimension of a and b

101. Consider two physical quantities A and B related to each other as $E = \frac{B - x^2}{At}$ where Ex and t have dimensions of energy, length and time respectively. The dimension of (AB) is

(a) $L^{-2}M^{1}T^{0}$ (b) $L^{2}M^{-1}T^{1}$ (c) $L^{0}M^{-1}T^{-1}$ (d) $L^{-2}M^{-1}T^{1}$

102. $Y = \log e^{\alpha t}$ then find dimension of α?

103. $F = 2V - 6t$. Find dimension of '2' & '6' Where F is force, V = velocity, t = time

(a) Both are dimensionless (b) MT^{-1}, MLT^{-3}
(c) $ML^{-1}T^{-2}$, MLT^{-3} (d) MTT^{-3}, MT^{-1}

104. Force (F) and density (d) are related as $F = \frac{\alpha}{\beta + \sqrt{d}}$ Then, the dimensions of α and β are

(a) $[M^{3/2}\,L^{-1/2}\,T^{-2}]$, $[ML^{-3}\,T^{0}]$
(b) $[M^{3/2}\,L^{-1/2}\,T^{-2}]$, $[M^{1/2}\,L^{-3/2}T^{0}]$
(c) $[M^{2}\,L^{2}\,T^{-1}]$, $[ML^{-1}\,T^{-3/2}]$
(d) $[MLT^{-2}]$, $[ML^{-2}\,T^{-2/3}]$

105. **Statement-1:** The displacement cannot be added with distance travelled.

Statement-2: Addition of scalar and vector of same dimensions is not a meaningful algebraic operation.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True

106. The force is given in terms of time t and displacement x by the equation $F = A\cos Bx + C\sin Dt$

The dimensional formula of $\frac{AD}{B}$ is

(a) $[M^{2}L^{2}T^{-3}]$ (b) $[M^{1}L^{1}T^{-2}]$ (c) $[ML^{2}T^{-3}]$ (d) $[M^{0}LT^{-1}]$

Dimensional Analysis

107. If time (t), energy (E) and momentum (P) taken as base quantities then dimension of mass (m), is

(a) $m = k\,t^{0}\,P^{2}\,E^{-1}$ (b) $m = k\,t^{0}\,P^{-1}\,E^{2}$
(c) $m = k\,t\,P^{2}\,E$ (d) $m = k\,t^{0}\,P^{2}\,E$

108. If mass (M), velocity (V) and time (T) are taken as fundamental units, then the dimensions of force (F) are

(a) [M V T] (b) $[M\,V\,T^{-1}]$
(c) $[M^{2}\,V\,T]$ (d) $[M^{-1}V^{-1}T]$

109. If force (F), acceleration (a) and time t is used as a fundamental P . Q then find dimension of Area in terms of them :-

(a) $F^{0}\,a^{1}\,T^{2}$ (b) $F\,a^{2}\,T^{2}$
(c) $F^{0}\,a^{2}\,T^{4}$ (d) $F^{0}\,a\,T$

110. The frequency of vibrations f of a mass m suspended from a spring of spring constant K is given by a relation of type $f = cm^{x}\,K^{y}$, where c is a dimensionless constant. The values of x and y are

(a) $x = \frac{1}{2}, y = \frac{1}{2}$ (b) $x = \frac{-1}{2}, y = \frac{-1}{2}$
(c) $x = \frac{1}{2}, y = \frac{-1}{2}$ (d) $x = \frac{-1}{2}, y = \frac{1}{2}$

111. If mass is written as $m = k\,c\,{}^{P}G^{-1/2}h^{1/2}$ then the value of P will be : (Constants have their usual meaning with k a dimensionless constant)

(a) 1/2 (b) 1/3 (c) 2 (d) –1/3

112. If force acceleration and time are basic fundamental P.Q. then find dimension of energy

(a) $F^{2}A^{-1}T$ (b) $F\,A\,T^{2}$
(c) $F\,A\,T^{-2}$ (d) $F\,A^{-1}T$

113. The frequency (v) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as: $v = r^{a}\rho^{b}s^{c}$. The values of a, b and c respectively are

(a) $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$ (b) $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$
(c) $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ (d) $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

114. The speed of a wave produced in H_2O is given by $v = \lambda^{a}\,g^{b}\,\rho^{c}$, where λ, g, ρ pare wavelength of wave, acceleration due to gravity & density of water respectively. The value of a, b & c respectively are :

(a) $\frac{1}{2}, 0, \frac{1}{2}$ (b) 1, 1, 0 (c) 1, –1, 0 (d) $\frac{1}{2}, \frac{1}{2}, 0$

115. If force F, area A and density D are taken as the fundamental units, the representation of Young's modulus 'Y' will be:

(a) $[F^{-1}A^{-1}D^{-1}]$ (b) $[FA^{-2}D^{2}]$
(c) $[FA^{-1}D]$ (d) $[FA^{-1}D^{0}]$

116. If maximum acceleration of oscillating particle is α and maximum velocity is β, then find time period

(a) $2\pi\frac{\alpha}{\beta}$ (b) $2\pi\alpha\beta$ (c) $2\pi\frac{\beta}{\alpha}$ (d) $\frac{2\pi\alpha^{2}}{\beta^{2}}$

117. Plank's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants, Which of the following combinations of these has the dimension of length?

(a) $\frac{\sqrt{hG}}{c^{3/2}}$ (b) $\frac{\sqrt{hG}}{c^{5/2}}$ (c) $\sqrt{\frac{hc}{G}}$ (d) $\frac{\sqrt{Gc}}{h^{3/2}}$

118. If dimensions of critical velocity v_c of a liquid flowing through a tube are expressed as $\eta^{x}\,\rho^{y}\,r^{z}$ where $\eta\ \rho\ r$ are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x, y and z are given by

(a) 1,1,1 (b) 1, –1,–1 (c) –1,–1,1 (d) –1,–1,–1

119. If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is:

(a) $[PA^{-1}T^{0}]$ (b) $[PAT^{-1}]$ (c) $[PA^{-1}T]$ (d) $[PA^{-1}T^{-1}]$

120. A spherical ball is moving through a viscous medium. If the viscous force acting on the ball is proportional to speed of the ball, then the dimensions of proportionality constant is

(*a*) $[M\,L^{-1}\,T^{-1}]$ (*b*) $[M\,L^{-2}\,T^{-1}]$

(*c*) $[M\,L\,T^{-2}]$ (*d*) $[M\,L^{0}\,T^{-1}]$

121. Force acting on object is proportional to square of velocity then find dimensions of proportional constant.

122. If energy E, velocity *V* and time T are taken as fundamental units, the dimensional formula for surface tension is

(*a*) $[E\,V^{-2}\,T^{-2}]$ (*b*) $[E^{-2}\,V\,T^{-2}]$

(*c*) $[E^{-2}\,V\,T^{-2}]$ (*d*) $[E^{-2}\,V^{-2}\,T^{2}]$

123. A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is c. is velocity of light, G is universal constant of gravitation and θ is charge]

(*a*) $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$ (*b*) $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

(*c*) $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{1/2}$ (*d*) $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$

124. Car is moving on rough inclined plane of inclination θ and coefficient of friction μ, then find maximum safe velocity of car?

125. The speed of light C gravitational constant G and Plank's constant h are taken as fundamental P.Q then the dimension of time in the new system of unit

(*a*) $G^{+1/2}\,h^{1/2}\,C^{-5/2}$ (*b*) $G^{-1/2}\,h^{1/2}\,C^{-1/2}$

(*c*) $G^{1/2}\,h^{1/2}\,C^{3/2}$ (*d*) $G^{-1/2}\,h^{-1/2}\,C^{-1/2}$

126. In a new system of units energy (E), density (d) and power (P) are taken as fundamental units, then the dimensional formula of universal gravitational constant G will be

(*a*) $[E^{-1}d^{-2}P^{2}]$ (*b*) $[E^{-1}d^{-1}P^{2}]$

(*c*) $[E^{2}\,d^{-1}\,P^{-1}]$ (*d*) $[E^{1}\,d^{-2}\,P^{-2}]$

127. A highly rigid cubical block A of small mass M and side L is fixed onto another cubical block B of the same dimensions and of low modulus of rigidity *n* such that the lower face of A completely covers the upper face of B. The lower face B is rigidly held on a horizontal surface. A small force F is applied perpendicular to one of the side faces of A. After the force is withdrawn, block A executes small oscillations. The time period of which is given by

(*a*) $2\pi\sqrt{\frac{M\eta}{L}}$ (*b*) $2\pi\sqrt{\frac{L}{M\eta}}$ (*c*) $2\pi\sqrt{\frac{ML}{\eta}}$ (*d*) $2\pi\sqrt{\frac{M}{\eta L}}$

128. Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity *x* as follows: [position] = $[x^{\alpha}]$; speed = $[x^{\beta}]$; [acceleration] = $[x^{p}]$; [linear momentum] = $[x^{q}]$; force = $[x^{r}]$. Then

(*a*) $\alpha + p = 2\beta$ (*b*) $p + q - r = \beta$

(*c*) $p - q + r = \alpha$ (*d*) $p + q + r = \beta$

129. Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of *L*, which of the following statement (*s*) is/are correct?

(*a*) The dimension of force is L^{-3}

(*b*) The dimension of energy of L^{-2}

(*c*) The dimension of power is L^{-5}

(*d*) The dimension of linear momentum is L^{-1}

130. A length-scale (*l*) depends on the permittivity (ε) of a dielectric material, Boltzmann constant (k_B), the absolute temperature (*T*), the number per unit volume (*n*) of certain charged particles, and the charge (*q*) carried by each of the particles. Which of the following expressions for *l* is (are) dimensionally correct?

(*a*) $l = \sqrt{\left(\frac{nq^2}{\varepsilon k_B T}\right)}$ (*b*) $l = \sqrt{\left(\frac{\varepsilon k_B T}{nq^2}\right)}$

(*c*) $l = \sqrt{\left(\frac{q^2}{\varepsilon n^{2/3} k_B T}\right)}$ (*d*) $l = \sqrt{\left(\frac{q^2}{\varepsilon n^{1/3} k_B T}\right)}$

131. Young's modulus of elasticity *Y* is expressed in terms of three derived quantities, namely, the gravitational constant *G*. Planck's constant *h* and the speed of light *c*, as $Y = c^{\alpha}h^{\beta}G^{\gamma}$ Which of the following is the correct option?

(*a*) $\alpha = 7, \beta = -1, \gamma = -2$ (*b*) $\alpha = -7, \beta = -1, \gamma = -2$

(*c*) $\alpha = 7, \beta = -1, \gamma = -2$ (*d*) $\alpha = -7, \beta = 1, \gamma = -2$

132. If momentum (p), area (A) and time (T) are taken to be fundamental quantities, then energy has the dimensional formula.

(*a*) $[pA^{-1}T^{1}]$ (*b*) $[p^{2}AT]$

(*c*) $[pA^{-1/2}T]$ (*d*) $[pA^{1/2}T^{-1}]$

133. If surface tension (*S*), Moment of inertia (*l*) and Planck's constant (*h*), were to be taken as the fundamental units, the dimensional formula for linear momentum would be.

(*a*) $S^{3/2}I^{1/2}h^{0}$ (*b*) $S^{1/2}I^{1/2}h^{0}$

(*c*) $S^{1/2}I^{1/2}h^{-1}$ (*d*) $S^{1/2}I^{3/2}h^{-1}$

134. If force (F), velocity (V) and time (T) are considered as fundamental physical quantity, then dimensional formula of density will be:

(*a*) $FV^{-2}T^{2}$ (*b*) $FV^{-4}T^{-2}$

(*c*) $FV^{4}T^{-6}$ (*d*) $F^{2}V^{2}T^{2}$

135. Velocity (*v*) and acceleration (*a*) in two systems of units 1 and 2 are related as respectively. Here *m* and *n* are constants. The relations for distance and time in two systems respectively are:

(*a*) $\frac{n^3}{m^3}L_1 = L_2$ and $\frac{n^2}{m}T_1 = T_2$

(*b*) $L_1 = \frac{n^4}{m^2}L_2$ and $T_1 = \frac{n^2}{m}T_2$

(*c*) $L_1 = \frac{n^2}{m}L_2$ and $T_1 = \frac{n^4}{m^2}T_2$

(*d*) $\frac{n^2}{m}L_1 = L_2$ and $\frac{n^4}{m^2}T_1 = T_2$

136. Consider the efficiency of Carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta}\log_e\frac{\beta x}{kT}$ where α and β are constants. If *T* is temperature, *k* is Boltzman constant, θ is angular displacement and *x* has the dimensions of length. Then, choose the incorrect option

(*a*) Dimensions of β is same as that of force

(*b*) Dimensions of $\alpha^{-1}x$ is same as that of energy

(*c*) Dimensions of $\eta^{-1}\sin\theta$ theta is same as that of αβ

(*d*) Dimensions of α is same as that of β

137. If E and H represents the intensity of electric field and magnetising field respectively, then the unit of E/H will be:

(a) Joule (b) Newton

(c) Ohm (d) Mho

138. A physical quantity $\vec{S}$ is defined as $\vec{S} = \left(\vec{E} \times \vec{B}\right) / \mu_0$, where $\vec{E}$ is electric field, $\vec{B}$ is magnetic field and μ_0 is the permeability of free space. The dimensions of $\vec{S}$ are the same as the dimensions of which of the following quantity (ies)?

(a) $\dfrac{\text{Energy}}{\text{Charge} \times \text{ Current}}$ (b) $\dfrac{\text{Force}}{\text{Length} \times \text{Time}}$

(c) $\dfrac{\text{Energy}}{\text{Volume}}$ (d) $\dfrac{\text{Power}}{\text{Area}}$

139. Identify the pair of physical quantities that have same dimensions:

(a) Velocity gradient and decay constant

(b) Wien's constant and Stefan constant

(c) Angular frequency and angular momentum

(d) Wave number and Avogadro number

140. Which of the following physical quantities have the same dimensions?

(a) Electric displacement ($\vec{D}$) and surface charge density

(b) Displacement current and electric field

(c) Current density and surface charge density

(d) Electric potential and energy

141. Which of the following is not a dimensionless quantity?

(a) Relative magnetic permeability (μr)

(b) Power factor

(c) Quality factor

(d) Permeability of free space (μ_0)

142. Which of the following combinations has the dimension of electrical resistance (Σ_0 is the permittivity of vacuum and μ_0 is the permeability of vacuum)?

(a) $\sqrt{\dfrac{\epsilon_0}{\mu_0}}$ (b) $\dfrac{\mu_0}{\epsilon_0}$ (c) $\dfrac{\epsilon_0}{\mu_0}$ (d) $\sqrt{\dfrac{\mu_0}{\epsilon_0}}$

143. In a typical combustion engine the work done by a gas molecule is given by $W = \alpha^2 \beta e^{-\frac{\beta x^2}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If a and b are constants, dimensions of a will be :

(a) $[MLT^{-1}]$ (b) $[M^0 L T^0]$

(c) $[M^2LT^{-2}]$ (d) $[MLT^{-2}]$

144. Dimensional formula for thermal conductivity is (here K denotes the temperature):

(a) $MLT^{-3} K^{-1}$ (b) $MLT^{-2} K^{-2}$

(c) MLT–2 K (d) $MLT^{-3} K$

145. Amount of solar energy received on the earth's surface per unit area per unit time is defined a solar constant. Dimension of solar constant is

(a) ML^2T^{-2} (b) MLT^{-2}

(c) $M^2L^0T^{-1}$ (d) ML^0T^{-3}

Limitation of Dimensional Analysis

146. Which of the following equation can be derived dimensionally??

(a) $s = vt - \dfrac{1}{2}at^2$ (b) $v^2 = u^2 \times 2as$

(c) $h = \dfrac{\omega^2 r^2}{2g}$ (d) $v = \dfrac{d}{t}$

147. The method of dimensions cannot be applied to derive the formula for physical quantity which depends on

(a) Two different physical quantities

(b) Three physical quantities in which two have same dimensions

(c) More than three physical quantities

(d) Both (b) & (c)

148. Which of the following equation can not be derived dimensionally

(a) $F = 6\pi r\eta$ (b) $\theta = \omega t$

(c) $\dfrac{d\theta}{dt} = \rho A v$ (d) $P = \rho gh$

149. Fill in the blanks with correct statement, according to given statement

Physically correctness	(1) _____	(2) _____	(c) Equation is physically wrong	(d) Equation is physically correct
Dimensional correctness	(a) Equation is dimensional wrong	(b) Equation is dimensional correct	(3) _____	(4) _____

150. A student when discussing the properties of medium writes velocity of light in vacuum = velocity of light in medium. This formula is

(a) Dimensionally correct (b) Dimensionally incorrect

(c) Numerically incorrect (d) Both (a) and (c)

151. **Assertion (A):** Work = Torque is dimensionally correct but not physically.

Reason (R): Dimensional correctness of an equation ensures its physical correctness)

(a) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).

(b) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).

(c) If Assertion (A) is True but the Reason (R) is False.

(d) If both Assertion (A) & Reason (R) are false.

Significant Digits

152. Give the number of significant figures in each measurement.

(a) 60.36.7 m (b) 0.006606 s

(c) 61.2.002 kg (d) 306,490,000 people

153. The number of significant figures in 3.04×10^{23} is:

(a) 2 (b) 3 (c) 23 (d) 25

154. The number of significant figures in 0.01020 is:

(a) 1 (b) 2 (c) 3 (d) 4

155. Match the list -I with List-II

List-I		List-II	
(I)	1001	(A)	3
(II)	010.1	(B)	4
(III)	100.100	(C)	5
(IV)	0.0010010	(D)	6

156. Given $P = 0.0030\ m$, $Q = 2.40$ m and $R = 3000\ m$, the number of significant figures in P,Q,R are respectively :

(*a*) 1, 2, 1 (*b*) 2, 3, 1

(*c*) 4, 2, 1 (*d*) 4, 2, 4

157. How many significant figures should the answer to this calculation contain? $= \dfrac{1.014 + 0.07}{5.11}$

(*a*) 2 (*b*) 3 (*c*) 4 (*d*) 1

158. Add these three length : $l_1 = 0.307$m, 0.52 m and 0.4 m

(*a*) 1.22 m (*b*) 1.2 m

(*c*) 1.3 m (*d*) 1.7 m

159. Calculate the correct number of significant figures for the final solution:

(*i*) Round 4.7475 to 4 significant figures

(*ii*) Round 4.7465 to 4 significant figures

160. (i) $76.4 \times 180.4 = ?$

(ii) $934.9 \div 0.00455 = ?$

161. Number of significant digit in the result of $\dfrac{4.327}{2.515}$ m .

(*a*) 2 (*b*) 4

(*c*) 3 (*d*) 5

162. Taking into account of significant digit. What is the value of 9.99 m – 0.0099 m.

(*a*) 9.98 m (*b*) 9.890 m

(*c*) 9.9 m (*d*) 9.9801 m

163. The area of rectangle of length 55.3 m and breadth 25 m .

(*a*) 1382 (*b*) 1382.5

(*c*) 14×10^2 (*d*) 138×10^1

164. If A = 2.413 and B = 1.2 then find A + B

165. If = 25.5 and B = 5 then find $\dfrac{A}{B}$

166. Express the final answer to the proper number of significant figures.

(i) $101.2 + 18.702 = ?$

(ii) $202.88 - 1.013 = ?$

167. $\dfrac{1.5}{1.5}$ is equal to

(*a*) 1 (*b*) 1.0 (*c*) 1.00 (*d*) 1.000

168. When 96.54 is divided by 2.40, the correct result is:

(*a*) 40.2250 (*b*) 4.0225

(*c*) 40.23 (*d*) 40.2

169. Find round off value of x = 16.351

(*a*) 16 (*b*) 16.33 (*c*) 16.3 (*d*) 16.4

170. Find round off value of x = 3.250

(*a*) 3.2 (*b*) 3.25 (*c*) 3.5 (*d*) 3

171. If true value of length is 6.57 m then which of the following reading is most accurate.

(*a*) 6.52 m (*b*) 6.61 m

(*c*) 6.513 m (*d*) 6.68 m

172. Which of the following reading is most accurate?

(*a*) 2.4 m (*b*) 2.41 m (*c*) 2 m (*d*) 2.413 m

173. For greater accuracy, the quantity with higher power should have least error.

(*a*) True (*b*) False

174. The mean length of an object is 5 cm. Which of the following measurements is most accurate?

(*a*) 4.9 cm (*b*) 4.805 cm

(*c*) 5.25 cm (*d*) 5.4 cm

175. The length, breadth and thickness of a rectangular sheet of metal are 4.234m, 1.005 m and 2.01 cm, respectively. Given the area and volume of the sheet to correct significant figures.

(*a*) 8.7 m^2, 0.092 m^3 (*b*) 8.4 m^2, 0.092 m^3

(*c*) 9.2 m^2, 0.001^3 (*d*) 8.72 m^2, 0.0855 m^3

Error Analysis

176. Smaller the least count, higher is the accuracy of measurement.

(*a*) True (*b*) False

177. Absolute error of the measurement is defined as

(*a*) The magnitude of the difference between individual and the true value of quantity

(*b*) The difference between individual and the true value

(*c*) Positive or negative value

(*d*) Both (*b*) and (*c*)

178. Absolute error is not dimensionless quantity.

(*a*) True (*b*) False

179. In a series of successive measurements in an experiment, the readings of the period of oscillation of a simple pendulum were found to be 2.63s, 2.56s, 2.42s, 2.71s and 2.80s. Calculate

(i) The mean value of the period of oscillation

(ii) The absolute error in each measurement

(iii) The mean absolute error

(iv) The relative error

(v) The percentage error

180. **Assertion (A):** $T_1 = (104° \pm 0.3)°C$; $T_2 = (100 \pm 0.1)°C$. $T_1 - T_2 = (4 \pm 0.4)°C$; $T_1 + T_2 = (204 \pm 0.4)°C$

Reason (R): Mean absolute error is added always but not applicable for absolute value.

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

181. Kallua measure mass of object as m = 4.95 kg.Where True mass is 5kg.

182. Measured a length of rod l1 = 5.02 cm $l_2 = 5.0$ 4 cm $rl_3 = 4.94$ cm then find mean absolute and percentage error in measurement.

183. Relative error is a dimensionless quantity.

(*a*) True (*b*) False

184. Absolute error and actual value of P.Q are 5 and 15 then relative and % error.

185. A wire has a mass (0.3 ± 0.003) g, radius (0.5 ± 0.005) mm and length (6 ± 0.06) cm. The maximum percentage error in the measurement of its density is:

(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4

186. If $A = (6 \pm 1)$ and $B = (4 \pm 2)$ m then find $Z = 2A + B$ with absolute error.

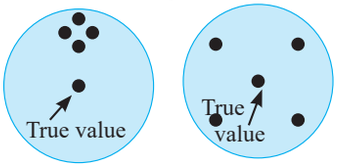
187. In an experiment to measure the height of a bridge by dropping stone into water underneath, if the error in measurement of time is 0.1 s at the end of 2 s, then the error in estimation of height of bridge will be

(*a*) 0.49 m (*b*) 0.98 m

(*c*) 1.96 m (*d*) 2.12 m

188. In an experiment to find acceleration due to gravity (*g*) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillation with a watch of 1s resolution. If measured value of length is 10 cm known to 1 mm accuracy. The accuracy in the determination of *g* is found to be *x*%. The value of x is

189.



Assertion (A): Instrument (i) is less accurate but more precise.

Reason (R): We will prefer instrument (i) for measurement.

(*a*) Both Assertion (A) and Reason (R) are correct and Reason (R) is correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are correct and Reason (R) is not correct explanation of Assertion (A).

(*c*) Assertion (A) is correct but Reason (R) is wrong.

(*d*) Both Assertion (A) and Reason (R) are wrong.

Relative Error

190. Value of $A = (10 \pm 0.1)$ m and $B = (20 \pm 0.5)$ m then find:

(*a*) $x = B + A$ (*b*) $y = B - A$

(*c*) $z = B.A$ (*d*) $m = B/A$

191. A public park, in the form of a square, has an area of (100 ± 0.2) m^2. The side of park is

(*a*) (10 ± 0.01) m (*b*) (10 ± 0.1) m

(*c*) (10 ± 0.02) m (*d*) (10 ± 0.2) m

192. If Z = A + B, find % error in Z.

193. If $Z = \frac{A}{B}$, Find absolute error in Z.

194. The external and internal diameters of a hollow cylinder are measured to be (4.23 ± 0.01) cm and (3.89 ± 0.01) cm. The thickness of the wall of the cylinder is:

(*a*) (0.34 ± 0.02) cm (*b*) (0.17 ± 0.02) cm

(*c*) (0.17 ± 0.01) cm (*d*) (0.34 ± 0.01) cm

195. The temperature of two bodies measured are $\theta_1 = 10°C \pm 0.4°C$ and $\theta_2 = 40°C \pm 0.3°C$. Find the sum and difference in temprature with error limit.

196. Percentage error in measuring the radius and mass of a solid sphere are 2% & 1% respectively. Then error in measurement of moment of inertia with respect to its diameter is :-

(*a*) 3% (*b*) 6% (*c*) 5% (*d*) 4%

197. $Z = \frac{4A^{\frac{1}{2}}\sqrt{B}}{C^{\frac{1}{3}}D}$, Find % error in Z

198. In an experiment four quantities *a*, *b*, *c* and *d* are measured with percentage error 1%, 2%, 3% and 4%, respectively. Quantity *P* is calculated as follows: $P = \frac{a^3b^2}{cd}$, % error in *P* is

(*a*) 10% (*b*) 7% (*c*) 4% (*d*) 14%

199. If the error in the measurement of radius of a sphere is 2%, then the error in the determination of volume of the sphere will be:

(*a*) 2% (*b*) 4% (*c*) 6% (*d*) 8%

200. The error in measuring the mass and radius of a solid sphere are ±1% and ±0.5% respectively. The error in calculation of the moment of inertia of the solid sphere will be about:

(*a*) ±1.5% (*b*) +2.5% (*c*) ±3% (*d*) ±2

201. If error in measuring diameter of a circle is 4%, the error in circumference of the circle would be:

(*a*) 2% (*b*) 8% (*c*) 4% (*d*) 1%

202. A force *F* is applied on a square area of side *L*. If the percentage error in the measurement of *L* is 2% and that in *F* is 4%, what is the maximum percentage error in pressure?

(*a*) 2% (*b*) 4% (*c*) 6% (*d*) 8%

203. If the percentage error in the measurement of momentum and mass of an object are 2% and 3% respectively, then maximum percentage error in the calculate value of its kinetic energy is

(*a*) 2% (*b*) 1% (*c*) 5% (*d*) 7%

204. A rectangular plate has length (2 + 0.02) cm and width (1 ± 0.01) cm. The maximum percentage error in the measurement of its area is

(*a*) 1% (*b*) 2% (*c*) 3% (*d*) 5%

205. The radius of a spherical body is (10 ± 0.5) cm. Then correct option is/are

(*a*) The percentage error in radius is 5%

(*b*) The percentage error in volume is 10%

(*c*) The percentage error in volume is 15%

(*d*) Both (*a*) & (*c*)

206. The least count of a stop watch is 1/5 second. The time of 20 oscillations of a pendulum is measured to be 25 seconds. The maximum percentage error in the measurement of time will be

(*a*) 0.1% (*b*) 0.8% (*c*) 1.8% (*d*) 8%

207. A physical quantity is represented by $X = [M^a L^b T^{-c}]$. If percentage error in the measurement of M, L and T are α%, β% and γ% respectively, then maximum percentage error in measurement of X should be (Given that α, β and γ are very small)

(*a*) $(\alpha a - \beta b + \gamma c)\%$ (*b*) $(\alpha a + \beta b + \gamma c)\%$

(*c*) $(\alpha a - \beta b - \gamma c)\%$ (*d*) $(\alpha a + \beta b - \gamma c)\%$

208. The acceleration due to gravity is measured on the surface of earth by using a simple pendulum. If α and β are relative errors in the measurement of length and time period respectively, then percentage error in the measurement of acceleration due to gravity is

(*a*) $\left(\alpha+\frac{1}{2}\beta\right)$ (*b*) $(\alpha-2\beta)$

(*c*) $(2\alpha+\beta)\times 100$ (*d*) $(\alpha+2\beta)\times 100$

209. Time period of simple pendulum $T=2\pi\sqrt{\frac{l}{g}}$, l is about 100 cm and is known to have 1 mm accuracy. Time period is 2s. The time of 100 oscillation measured by stop watch of L.C. 0.1.S. percentage error in g.

210. The resistance of a conductor R = V/I where V = (50 ± 2) volt and I = (9 ± 0.3) Amp, find percentage error in R. also find absoule error in R.

211. The radius of a sphere is (5.3 ± 0.1) cm. The percentage error in its volume is

(*a*) $\frac{0.1}{5.3}\times 100$ (*b*) $3\times\frac{0.1}{5.3}\times 100$

(*c*) $\frac{3}{2}\times\frac{0.1}{5.3}\times 100$ (*d*) $6\times\frac{0.1}{0.3}\times 100$

212. The length of a cylinder is measured with a meter rod having least count 0.1 cm. Its diameter is measured with Vernier caliper's having least count 0.01 cm. Given that length is 5 cm and radius is 2 cm. The percentage error in the calculated value of volume will be

(*a*) 1% (*b*) 2% (*c*) 3% (*d*) 4%

213. The relative error in the determination of the surface area of sphere is a. Then the relative error in the determination of its volume is:

(*a*) $\frac{3}{2}\alpha$ (*b*) $\frac{2}{3}\alpha$ (*c*) $\frac{5}{2}\alpha$ (*d*) α

214. $y = \sin\theta$ find percentage error in y if percentage error in θ is 2% at $\theta = \pi/6$ rad.

215. If $y = m\tan\theta$, where m is constant then find angle at which % error in 'y' will be minimum.

216. If percentage error in speed and mass are 1% and 2% then find % error in K.E.

217. Find percentage error in length of simple pendulum if % error in time is 4% and acceleration due to gravity is 2%

218. If error in measurement of speed is 50% then find % error in kinetic energy.

219. Two wire of resistance $R_1 = (50 \pm 2)\ \Omega$ and $R_2 = (100 \pm 4)\ \Omega$, find equivalent capacitance in (I) series (II) parallel with absolute error.

Types of Instruments & Least Count

220. If measured length of Rod is 1.56 cm then instrument used is

(*a*) metre scale (*b*) Vernier caliper

(*c*) screw gauge

221. Write down name of measuring instrument for given measurement:

(1) 87.3 mm (5) 8.53 cm

(2) 0.831 cm (6) 8.96 mm

(3) 6.7 cm (7) 9.812 cm

(4) 6.7 mm

222. A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it?

(*a*) A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.

(*b*) A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

(*c*) A meter scale.

(*d*) A Vernier caliper where the 10 divisions in Vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 cm.

223. Given below are two statements:

Statement I: Least count of a screw gauge is directly proportional to the number of divisions on circular scale.

Statement II: A screw gauge having a smaller value of pitch has greater accuracy.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both statement I and statement II are correct

(*b*) Statement I is correct and statement II is incorrect

(*c*) Statement I is incorrect and statement II is correct.

(*d*) Both statements I and statements II are incorrect

224. Match the quantities in column –I to dimensions in column-II. R is resistance, L is inductance, C is capacitance, H is latent heat and S is specific heat.

Column-I		Column-II	
(A)	LC	(p)	L^2T^{-2}
(B)	LR	(q)	$L^2T^{-2}K^{-1}$
(C)	H	(r)	T^2
(D)	S	(s)	$M^2L^4T^{-5}A^{-4}$

(*a*) A→r, B→s, C→p, D→q (*b*) A→s, B→p ,C→r, D→q

(*c*) A→p, B→s,C→r, D→q (*d*) A→q, B→s,C→r, D→q

Vernier Caliper

225. In Vernier calipers 8 M.S.D. coincides with 10 Vernier scale division then find least count.

226. Least count of main scale of Vernier caliper is 1 cm and 4 main scale division coincide with 6 Vernier scale division then L.C. of Vernier scale.

227. 10 divisions on the main scale of Vernier caliper coincide with x divisions on Vernier scale. If Each division on main scale is of 5 units and Vernier constant of given scale is 5/11, find the value of x.

(*a*) 11 (*b*) 12 (*c*) 9 (*d*) 13

228. In certain vernier callipers, 25 divisions on the vernier scale have the same length as 24 divisions on the main scale. One division on the main scale is 1 mm long. The least count of the instrument is

(*a*) 0.04 mm (*b*) 0.01 mm (*c*) 0.02 mm (*d*) 0.08 mm

229. In a Vernier calliper, one main scale division is x cm and n division of Vernier scale coincide with $(n-1)$ division of the main scale. The least count of the Vernier caliper in cm is:

(a) $\left(\frac{n-1}{n}\right)x$ (b) $\frac{nx}{(n-1)}$ (c) $\frac{x}{n}$ (d) $\frac{x}{n-1}$

230. The main scale of a Vernier callipers has n divisions/cm. n divisions of the Vernier scale coincide with $(n-1)$ divisions of main scale. The least count of the Vernier callipers is

(a) $\frac{1}{(n+1)(n-1)}$ cm (b) $\frac{1}{n}$ cm

(c) $\frac{1}{n^2}$ cm (d) $\frac{1}{n(n+1)}$ cm

231. One cm on the main scale of vernier callipers is divided into ten equal parts. If 20 divisions of vernier scale coincide with 8 small divisions of the main scale. What will be the least count of callipers?

232. A hypothetical Vernier scale of a travelling microscope has 40 divisions which is equal to 38 main scale divisions. If each main scale division is 1.2 mm, then minimum error in the measurement of length is

(a) 0.6 mm (b) 1.2 mm

(c) 0.06 mm (d) 0.1 mm

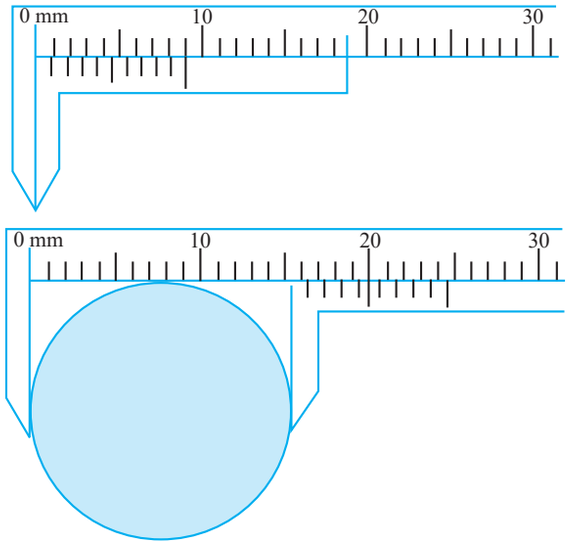
233. One centimeter on the main scale of Vernier calliper is divided into ten equal parts. If 10 divisions of Vernier scale coincide with 8 small divisions of the main scale, the least count of the callipers is:

(a) 0.01 cm (b) 0.02 cm

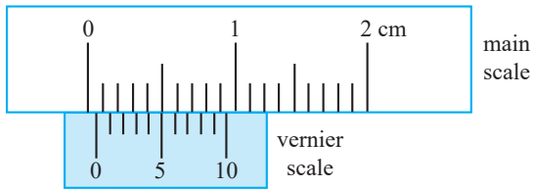
(c) 0.05 cm (d) 0.07 cm

Reading on Vernier Calliper

234. In the given Vernier scale, 10 divisions of the vernier scale is matching with 9 divisions of the main scale as shown in the first figure. Find the diameter of the object in the second figure. Assume the edge of the vernier as '0' of the verrnier.



235. Find the zero correction in the given figure.



(a) 0.4 mm (b) 0.5 mm

(c) –0.5 mm (d) –0.4 mm

236. The diameter of a cylinder is measured using a vernier calipers with no zero error. It is found that the zero of the vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is

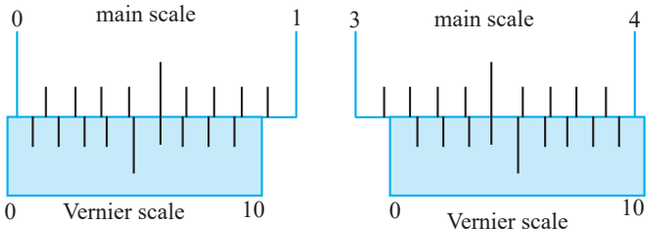
(a) 5.112 cm (b) 5.124 cm

(c) 5.136 cm (d) 5.148 cm

237. The least count of the main scale of a vernier calipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is (VSD is vernier scale division)

(a) 2.99 cm (b) 3.07 cm (c) 3.21 cm (d) 3.2 cm

238. The smallest division on the main scale of a Vernier callipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this calliper with no gap between its two jaws. The figure on the right shows the reading with a solid sphere held between the jaws. The correct diameter of the sphere is



(a) 3.07 cm (b) 3.11 cm (c) 3.15 cm (d) 3.17 cm

Screw Gauge

239. A screw gauge has least count of 0.01 mm and there arc 50 divisions in its circular scale. The pitch of the screw gauge is

(a) 0.01 mm (b) 0.25 mm (c) 0.5 mm (d) 1.0 mm

240. Using screw gauge of pitch 0.1 cm and 50 divisions on its circular scale, the thickness of an object is measured. It should correctly be recorded as

(a) 2,123 cm (b) 2.124 cm

(c) 2.125 cm (d) 2.121 cm

241. A screw gauge has 50 divisions on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5 mm is noticed on the pitch scale. The nature of zero error involved and the least count of the screw gauge, are respectively:

(*a*) Positive, 0.1 mm (*b*) Positive, 0.1 μm

(*c*) Positive, 10 μm (*d*) Negative, 2 μm

242. **Assertion (A):** If in five complete rotations of the circular scale, the distance travelled on main scale of the screw gauge is 5 mm and there are 50 total divisions on circular scale, then least count is 0.001 cm.

Reason (R): Least Count = $\frac{\text{Pitch}}{\text{Total divisions on circular sclae}}$

In the light of the above statements, choose the most appropriate answer from the options given below :

(*a*) Assertion (A) is not correct but Reason (R) is correct.

(*b*) Both Assertion (A) and Reason (R) are correct and Reason (R) is the correct explanation of Assertion (A).

(*c*) Assertion (A) is correct but Reason (R) is not correct.

(*d*) Both Assertion (A) and Reason (R) are correct and Reason (R) is NOT the correct explanation of Assertion (A)

Reading on Screw Guage

243. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of –0.004 cm, the correct diameter of the ball is

(*a*) 0.521 cm (*b*) 0.525 cm (*c*) 0.053 cm (*d*) 0.529 cm

244. Two full turns of the circular scale of gauge cover a distance of 1 mm on scale. The total number of divisions on circular scale is 50. Further, it is found that screw gauge has a zero error of -0.03 mm. While measuring the diameter of a thin wire a student notes the main scale reading of 3 mm and the number of circular scale division in line, with the main scale as 35. The diameter of the wire is

(*a*) 3.32 mm (*b*) 3.73 mm (*c*) 3.67 mm (*d*) 3.38 mm

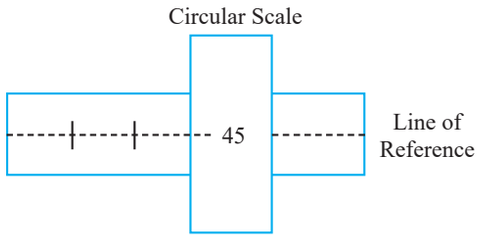
245. The pitch of a screw gauge is 1 mm and there are 100 division on its circular scale. When nothing is put in between its jaws, the zero of the circular scale lies 4 divisions below the reference line, When a steel wire is placed between the jaws, two main scale divisions are clearly visible and 67 divisions on the circular scale are observed. The diameter of the wire is

(*a*) 2.71 mm (*b*) 2.67 mm (*c*) 2.63 mm (*d*) 2.65 mm

246. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides the main scale line?

(*a*) 0.75 mm (*b*) 0.80 mm (*c*) 0.70 mm (*d*) 0.50 mm

247. Consider a screw gauge without any zero error. What will be the final reading corresponding to the final state as shown? It is given that the circular head translates P MSD in N rotations. One MSD is equal to 1 mm.



The circular scale has 100 divisions

(*a*) $\left(\frac{P}{N}\right)\left(2+\frac{45}{100}\right)$ mm (*b*) $\left(\frac{N}{P}\right)\left(2+\frac{45}{N}\right)$ mm

(*c*) $P\left(\frac{2}{N}+\frac{45}{100}\right)$ mm (*d*) $\left(2+\frac{45}{100}\times\frac{P}{N}\right)$ mm

248. There are 100 divisions on the circular scale of a screw gauge of pitch 1 mm. With no measuring quantity in between the jaws, the zero of the circular scale lies 5 divisions below the reference line. The diameter of a wire is then measured using this screw gauge. It is found the 4 linear scale divisions are clearly visible while 60 divisions on circular scale coincide with the reference line. The diameter of the wire is:

(*a*) 4.65 mm (*b*) 4.55 mm (*c*) 4.60 mm (*d*) 3.35 mm

Types of Error

249. Zero error of an instrument introduces:

(*a*) Systematic errors (*b*) Random errors

(*c*) Both (*d*) None

250. **Assertion (A):** In Vernier caliper, if positive zero error exists, then while taking measurements the reading taken will be more than actual Reading.

Reason (R): The Zero error in Vernier caliper might have happened due to manufacturing defect or due to rough handling.

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

251. In 5 number of observation systematic error is 12% then find error in 20 observation?

252. In 5 number of observation random error is 12% then find error in 20 observation?

253. Given below are two statements:

Statement I: Absolute error is unitless and dimensionless.

Statement II: All types of errors are unitless and dimensionless.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both statement I and statement II are correct

(*b*) Statement I is correct and statement II is incorrect.

(*c*) Statement I is incorrect and statement II is correct

(*d*) Both statements I and statements II are incorrect.

254. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): A spherical body of radius (5 ± 0.1) mm having a particular density is falling through a liquid of constant density. The percentage error in the calculation of its terminal velocity is 4%.

Reason (R): The terminal velocity of the spherical body falling through the liquid is inversely proportional to its radius. In the light of the above statements, choose the correct answer from the options given below

(*a*) If both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of the Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(*c*) If Assertion (A) is true, but Reason (R) is false

(*d*) If Assertion (A) is false, but Reason (R) is true.

(*e*) If both Assertion (A) and Reason (R) are false.

255. The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm respectively. What will be the value of its volume in appropriate significant figures?

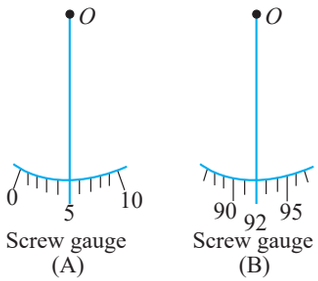
(*a*) $4264 \pm 81\ cm^3$

(*b*) $4264 \pm 81.0\ cm^3$

(*c*) $4260 \pm 80\ cm^3$

(*d*) $4300 \pm 80\ cm^3$

256. Student *A* and Student *B* used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students *A* and *B* is.....



Screw gauge (A) Screw gauge (B)

[Figure shows position of reference '*O*' when jaws of screw gauge are closed]

Given pitch = 0.1 cm.

MR* CORNER

1. **Assertion (A):** Force can be added to pressure.
   Reason (R): Force and pressure have same dimensions.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
2. **Assertion (A):** Both velocity and speed have same dimensions.
   Reason (R): Velocity cannot be added to speed.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
3. **Assertion (A):** The given equation $x = x_0 + u_0t + 1/2\ at^2$ is dimensionally correct, where x is the distance travelled by a particle in time t, initial position x_0 initial velocity u_0 and uniform acceleration a is along the direction of motion.
   Reason (R): Dimensional analysis can be used for checking the dimensional consistency or homogeneity of the equation.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
4. **Assertion (A):** Mass, length and time are fundamental physical quantities.
   Reason (R): They are independent of each other.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
5. **Assertion (A):** Density is a derived physical quantity.
   Reason (R): Density cannot be derived from the fundamental physical quantities.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
6. **Assertion (A):** When we change the unit of measurement of a quantity, its numerical value changes.
   Reason (R): Smaller the unit of measurement, smaller is its numerical value.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
7. **Assertion (A):** L/R and CR both have the same dimensions.
   Reason (R): L/R and CR both have the dimension of time.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
8. **Assertion (A):** A screw gauge having a smaller value of pitch has greater accuracy.
   Reason (R): The least count of screw gauge is directly proportional to the number of divisions on circular scale.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
9. **Assertion (A):** All unitless quantities are dimensionless.
   Reason (R): Dimensions are exponent raised to fundamental units in derived units.
   (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
   (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
   (*c*) If Assertion (A) is true but Reason (R) is false.
   (*d*) If both Assertion (A) and Reason (R) are false.
10. **Assertion (A):** Universal gravitation constant is a dimensional constant.
    Reason (R): $G = 6.67\times10^{-11}\ Nm^2\ kg^{-2}$.
    (*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
    (*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
    (*c*) If Assertion (A) is true but Reason (R) is false.
    (*d*) If both Assertion (A) and Reason (R) are false.
11. **Assertion (A):** Power of an engine depends on mass, angular speed, torque and angular momentum, so the formula of power is not derived with the help of dimensional method.

Reason (R): In mechanics, if a particular quantity depends on more than three quantities, then we cannot derive the formula of the quantity by the help of dimensional method.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

12. **Assertion (A):** The magnitude of any physical quantity does not depend on the method of measurement.

Reason (R): The magnitude of any physical quantity is more in SI system than in CGS system.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

13. **Assertion (A):** Temperature cannot be expressed as a derived quantity in terms of length and mass.

Reason (R): Temperature is a fundamental quantity.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

14. **Assertion (A):** Quality factor is dimensionless.

Reason (R): Quality factor depends on resistance, inductance and capacitance of LCR series circuit.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

15. **Assertion (A):** The unit of EMF is Joule/Coulomb.

Reason (R): EMF is an electromagnetic force.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

16. **Assertion (A):** A physical quantity is measured and its value is always found to be *nu*; where *n* is the numerical value and *u* is the unit.

Reason (R): $n \propto \frac{1}{u}$

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

17. **Assertion (A):** Accuracy of measurement is determined by absolute error and percentage error both.

Reason (R): Absolute error is the difference in actual and measured values of physical quantities.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

18. **Assertion (A):** The random error in the arithmetic mean of 100 observations is *x*; then random error in the arithmetic mean of 400 observations would be *x*/4.

Reason (R): Arithmetic mean of the magnitudes of absolute errors in *n* measurements of the quantity is represented by $\overline{\Delta a} = \frac{|\Delta a_1| + |\Delta a_2| + ... + |\Delta a_n|}{n}$

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

19. **Assertion (A):** The quantity percentage error is unitless.

Reason (R): Percentage error

$= \frac{\text{Mean absolute error}}{\text{Mean value}} \times 100\% = \frac{\overline{\Delta a}}{a_m} \times 100\%;$

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

20. **Assertion (A):** Systematic error can be minimised.

Reason (R): Systematic error can be calculated.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

21. **Assertion (A):** The period of oscillation of a simple pendulum in the experiment is recorded as 2.63 s, 2.56 s, 2.42 s, 2.71 s and 2.80 s, respectively. The average absolute error is 0.11 s.

Reason (R): Mean absolute error $= \frac{\text{Sum of absolute errors}}{\text{Numbers of observations}}$

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

22. **Assertion (A):** Measurement's precision is determined by least count of measuring instrument.

Reason (R): Smaller the least count, more is the precision

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

23. **Assertion (A):** Specific gravity of liquid is dimensionless.

Reason (R): It is the ratio of density of liquid to density of water.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false

24. **Assertion (A):** Methods of dimensions cannot be used for deriving formula containing trigonometric ratios.

Reason (R): Trigonometric ratios have no dimensions.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

25. **Assertion (A):** Both plane and solid angles are fundamental units.

Reason (R): Both have the same units.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

26. **Assertion (A):** Astronomical unit is a unit for measuring large distances.

Reason (R): It is the distance covered by light in one year.

(*a*) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) If Assertion (A) is true but Reason (R) is false.

(*d*) If both Assertion (A) and Reason (R) are false.

Answer Key

1. (*a*) 2. (*d*) 3. (*e*) 4. (*a*) 5. (*d*) 6. (*b*) 7. (*b*) 8. (*a*) 9. 2500 m/s

10. (*c*) 11. (*b*) 12. (*a*) 13. (*b*) 14. (*a*) 15. (*a*) 16. 5/8 17. (*c*) 18. (*b*) 19. (*d*)

20. (*c*) 21. (*b*) 22. (*a*) 23. (*d*) 24. (*d*) 25. (*d*) 26. (*c*) 27. (*c*) 28. Clock 2 (Presision is more)

29. (*d*) 30. (*a*) 31. (*c*) 32. (*a*) 33. $ML^{-1}T^{-2}$ 34. $ML^2T^{-3}A^{-2}$ 35. $ML^2T^{-2}A^{-2}$ 36. $MLT^{-2}A^{-2}$

37. $MT^{-2}A^{-1}$, $ML^2T^{-2}A^{-1}$ 38. $ML^{-1}T^{-1}$ 39. (*c*) 40. (*d*) 41. (*a*) 42. (*a*) 43. (*a*) 44. (*a*) 45. (*b*)

46. (*c*) 47. (*a*) 48. (*d*) 49. (*b*) 50. (*b*) 51. (*c*) 52. (*b*) 53. (*d*) 54. (*d*) 55. (*c*)

56. (*c*) 57. (*c*) 58. (*a*) 59. (*d*) 60. (*a*) 61. (*b*)

62. (1) May have dimension/may be dimensionless, (2) Must be dimensionless/does not have dimension, (3) Must have unit, (4) May or may not have unit 63. (*b*) 64. (*a*) 65. (*b*) 66. (*b*) 67. (*d*) 68. (*d*) 69. Yes 70. No 71. (*b*)

72. (*c*) 73. (*c*) 74. (*d*) 75. (*a*) 76. (*b*) 77. $a = MLT^{-3}$, $b = MLT^{-4}$ 78. $A = LT^{-1}$, $B = T^{-1}$

79. $A = MLT^{-2}$, $B = ML^2T^{-1}$, $C = t^2 = M^0L^0T^2$ 80. L^6 81. (*b*) 82. (*c*) 83. (*b*) 84. (*b*) 85. (*b*)

86. (*a*) 87. (*c*), (*d*) 88. $A = L^1$, $\omega = T^{-1}$, $K = L^{-1}$, $\phi = M^0L^0T^0$ 89. $\beta = L^{-1}$, $\alpha = T^{-1}$ 90. $\alpha = LT^{-2}$, $\beta = T^{-1}$, $\gamma = T^{-2}$

91. (*a*) 92. (*d*) 93. $[L^{-2}T^{-1}]$ 94. $\alpha = \left[M^{\frac{1}{2}}L^{-\frac{1}{2}}T^{-1}\right]$, $\beta = \left[M^{\frac{1}{2}}L^{-\frac{3}{2}}\right]$ 95. $a = [L^1] \cdot b = [M^0L^0T^0]$

96. $\alpha = [MLT^{-2}]$, $\beta = [M^0L^0T^0]$, $\gamma = [L^{-1}]$ 97. (*b*) 98. $A = L^1$, $k = L^{-1}$, $\omega = T^{-1}$ 99. $[LT^{-1}]$ 100. $b = M^{-1}$

101. (*b*) 102. $\alpha = T^{-1}$ 103. (*b*) 104. (*b*) 105. (*a*) 106. (*c*) 107. (*a*) 108. (*b*) 109. (*c*) 110. (*d*)

111. (*a*) 112. (*b*) 113. (*a*) 114. (*d*) 115. (*d*) 116. (*c*) 117. (*a*) 118. (*b*) 119. (*a*) 120. (*d*)

121. (*d*) 122. (*a*) 123. (*a*) 124. $v = \sqrt{Rg\dfrac{\tan\theta + \mu}{1 - \mu\tan\theta}}$ 125. (*a*) 126. (*b*) 127. (*d*) 128. (*a* & *b*) 129. (*a,b,d*)

130. (*b* & *d*) **131.** (*a*) **132.** (*d*) **133.** (*b*) **134.** (*b*) **135.** (*a*) **136.** (*d*) **137.** (*c*) **138.** (*b, d*) **139.** (*a*)

140. (*a*) **141.** (*d*) **142.** (*d*) **143.** (*b*) **144.** (*a*) **145.** (*d*) **146.** (*d*) **147.** (*d*) **148.** (*a*)

149. (1) Must be physically wrong, (2) May or amy not physically correct, (3) May or may be dimensionally correct

(4) Must be dimensionally correct $S_n = u + \frac{a}{2}(2n-1)$ (S_nth $\rightarrow$ dimensionally correct becouse it is displacement in one sec.) **149.** (*d*)

151. (*c*) **152.** 3, 4, 4, infinite. **153.** (*b*) **154.** (*d*) **155.** I-B, II-A, III-D, IV-C **156.** (*b*) **157.** (*b*) **158.** (*b*)

159. (i) 4.75, (ii) 4.75 **160.** 13700, 205000 **161.** (*b*) **162.** (*a*) **163.** (*c*) **164.** 3.6 **165.** 5

166. (i) 119.9 (ii) 201.87 **167.** (*b*) **168.** (*d*) **169.** (*d*) **170.** (*a*) **171.** (*b*) **172.** (*d*) **173.** (*a*) **174.** (*a*)

175. (*d*) **176.** (*a*) **177.** (*a*) **178.** (*a*) **179.** (i) 2.62, (ii) 0.01, 0.06, 0.2, 0.09, 0.18, (iii) 0.11, (iv) 0.042, (iv) 4.2%

180. (*a*) **181.** 1% **182.** 0.53% **183.** (*a*) **184.** 33.3% **185.** (*d*) **186.** [16 ± 4] **187.** (*c*) **188.** 5% **189.** (*b*)

190. (i) $\frac{\Delta x}{x} = \frac{0.6}{30}$, (ii) $\frac{\Delta y}{y} = \frac{0.6}{10}$, (iii) $\frac{\Delta z}{z} = \left(\frac{0.5}{20} + \frac{0.1}{10}\right)$, (iv) $\frac{\Delta m}{m} = \frac{7}{200}$ **191.** (*a*) **192.** $\frac{\Delta Z}{Z} \times 100 = \left(\frac{\Delta A + \Delta B}{A+B} \times 100\right)$

193. $(\Delta Z) = \frac{A}{B}\left(\frac{\Delta A}{A} + \frac{\Delta B}{B}\right)$ **194.** (*a*) **195.** sum = (50 ± 0.7)°C, Difference = (30 ± 0.7)°C **196.** (*c*)

197. $100 \times \frac{\Delta Z}{Z} = \left[\frac{1}{2}\frac{\Delta A}{A} + \frac{1}{2}\frac{\Delta B}{B} + \frac{1}{3}\frac{\Delta C}{C} + \frac{\Delta D}{D}\right] \times 100$ **198.** (*d*) **199.** (*c*) **200.** (*d*) **201.** (*c*) **202.** (*d*) **203.** (*d*)

204. (*b*) **205.** (*d*) **206.** (*b*) **207.** (*b*) **208.** (*d*) **209.** 10.1% **210.** 7.3%, $\frac{11}{27}$ **211.** (*b*) **212.** (*c*)

213. (*a*) **214.** $\frac{\pi}{\sqrt{3}}\%$ **215.** 45° **216.** 4% **217.** 10% **218.** 125% **219.** (I) (150 ± 6) Ω, (II) (33.1 ± 3.1) Ω **220.** (*b*)

221. (1) Vernier Calliper, (2) screw gauge, (3) metre scale, (4) Vernier Calliper, (5) Vernier Calliper, (6) screw gauge, (7) screw gauge 14

222. (*d*) **223.** (*c*) **224.** (*a*) **225.** [0.2 mm] **226.** 0.33 cm **227.** (*a*) **228.** (*a*) **229.** (*c*)

230. (*c*) **231.** 12/20 **232.** (*c*) **233.** (*b*) **234.** 15.6 mm **235.** (*b*) **236.** (*b*) **237.** (*b*) **238.** (*d*)

239. (*c*) **240.** (*b*) **241.** (*c*) **242.** (*a*) **243.** (*d*) **244.** (*d*) **245.** (*c*) **246.** (*b*) **247.** (*d*) **248.** (*b*)

249. (*a*) **250.** (*b*) **251.** 12% **252.** 3% **253.** (*d*) **254.** (*c*) **255.** (*c*) **256.** 0.013

MR* CORNER

1. (*d*) **2.** (*b*) **3.** (*a*) **4.** (*a*) **5.** (*c*) **6.** (*c*) **7.** (*a*) **8.** (*c*) **9.** (*b*) **10.** (*a*)

11. (*a*) **12.** (*d*) **13.** (*a*) **14.** (*b*) **15.** (*b*) **16.** (*a*) **17.** (*d*) **18.** (*b*) **19.** (*a*) **20.** (*d*)

21. (*a*) **22.** (*a*) **23.** (*a*) **24.** (*a*) **25.** (*d*) **26.** (*d*)